German University in Cairo

Reading Braille from a Phone Photo

Detecting and decoding every Braille cell on a page



b r a i l l e

Team 41

Youssef ElKhodary 58-18247

Yassin Mohamed 58-16061

Omar Tarek Hassan Fahmy 58-10021

Omar Gamaleldin 58-13583

Ahmed Tamer 58-14045

Supervisor: Dr. Mohamed Karam

Media Engineering and Technology (MET)

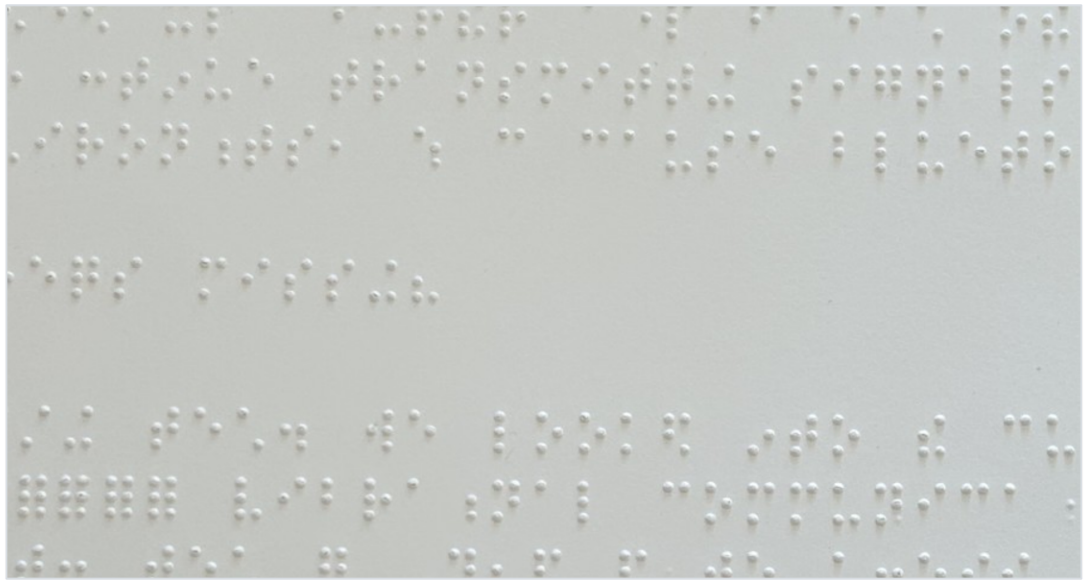
German University in Cairo

Agenda

1	Problem and motivation	1 min
2	**The data — sources, labels, challenges**	3½ min
3	Model, training strategy and tools	1 min
4	Evaluation plan	½ min

PROBLEM

Braille on paper is closed to sighted readers



What a sighted teacher sees: an English Braille page

43 M

blind people, 2020 [1]

< 10 %

of published works are accessible; < 1 % in poorer countries [2]

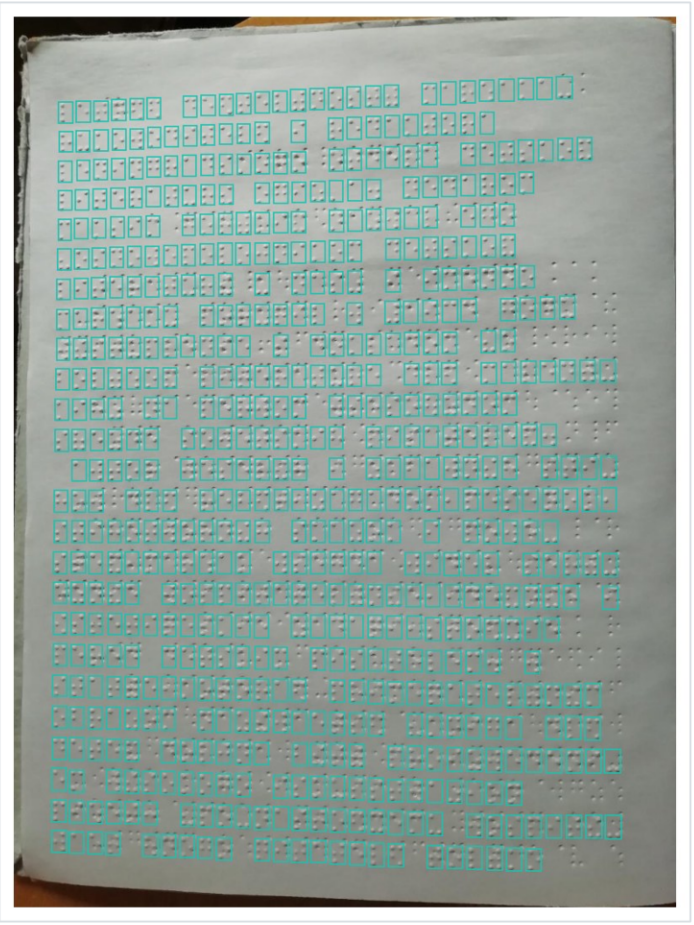
Today

- Teachers and parents decode homework dot by dot, by eye [3]
- Or wait for a Braille-literate transcriber
- Existing OBR tools expect flat scanner images

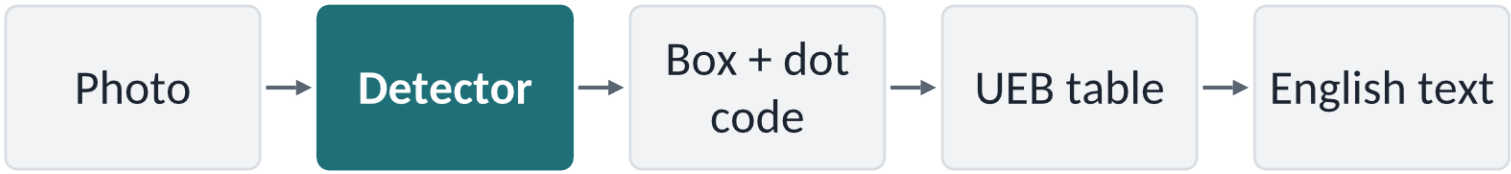
Goal: a phone photo of a Braille page → text, in seconds

[1] GBD 2019 Blindness & Vision Impairment Collaborators, Lancet Glob. Health 2021 [2] World Blind Union, via IAPB 2017 [3] Ovodov, ICCV Workshops 2021

The task: object detection with 63 classes



Input	One RGB photo of a Braille page (≈ 1024 × 1376 px)
Output	For each cell: a bounding box + 1 of 63 dot patterns
Why detection	≈ 294 cells per page, at unknown, distorted positions



Dataset 1: Angelina Braille Images Dataset

SOURCE
I. G. Ovodov, ICCV Workshops 2021

LINK
github.com/IlyaOvodov/AngelinaDataset

VERSION
v1.1 (May 2023)

LICENCE
None declared — cite the paper

FORMAT
JPEG photos + one CSV per photo

Contents	Train	Val	Test	Cells
Book pages (2-sided)	169	43	—	66,133
Handwritten sheets	22	6	—	6,117
Negatives (no Braille)	44	—	—	0
Real user uploads	—	—	50	13,013
Total: 334 photos	**235**	**49**	**50**	**85,263**

- 63 classes (dot patterns), ~294 cells per page
- 52 image sizes; 87 % are 1024 px wide
- Gives us: human-checked boxes to train and test on

Ovodov, “Optical Braille Recognition Using Object Detection Neural Network”, ICCV Workshops 2021, pp. 1741–1748

Dataset 2: English Braille Book Photos

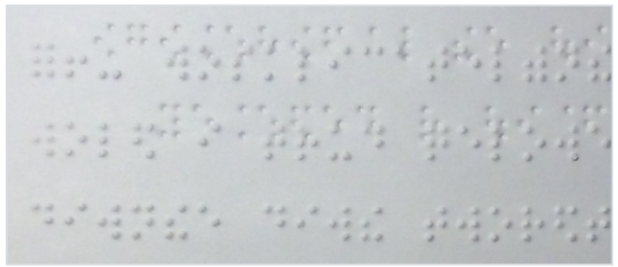
SOURCE
V. A. Zuev, Peter the Great St. Petersburg Polytechnic University, 2021

LINK
kaggle.com/datasets/valeryzuev/unlabeled-braille-photos

LICENCE
Not stated — cite, academic use

FORMAT
JPEG photos (1936 × 2592) + the book texts

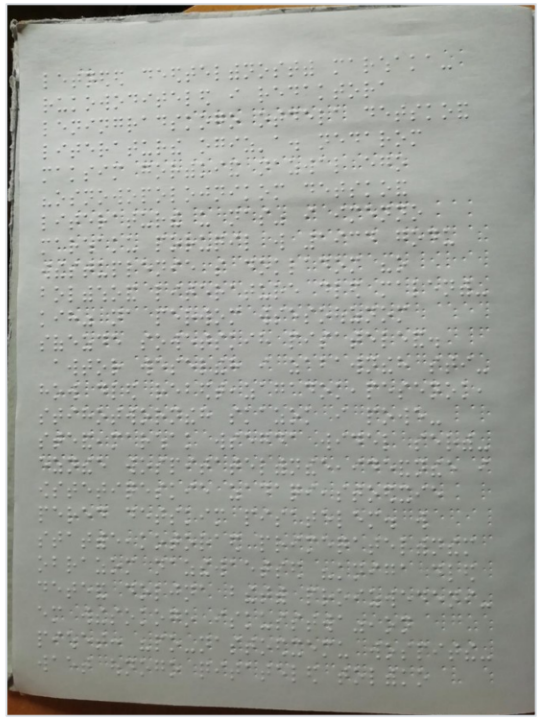
Contents	Volumes	Photos
Jane Eyre	8	1,193
The Scarlet Letter	4	538
Total (grade 1 English)	**12**	**1,731**



- Ground truth: the full text of both books (≈ 1.26 M Braille cells)
- Gives us: English pages; boxes come from Dataset 1

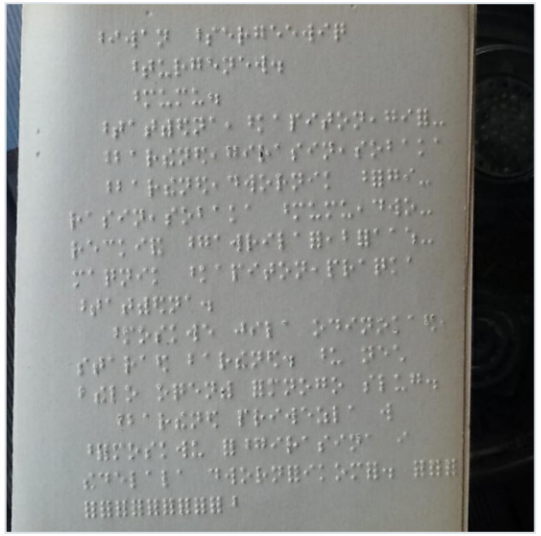
DATASET

What the photos look like



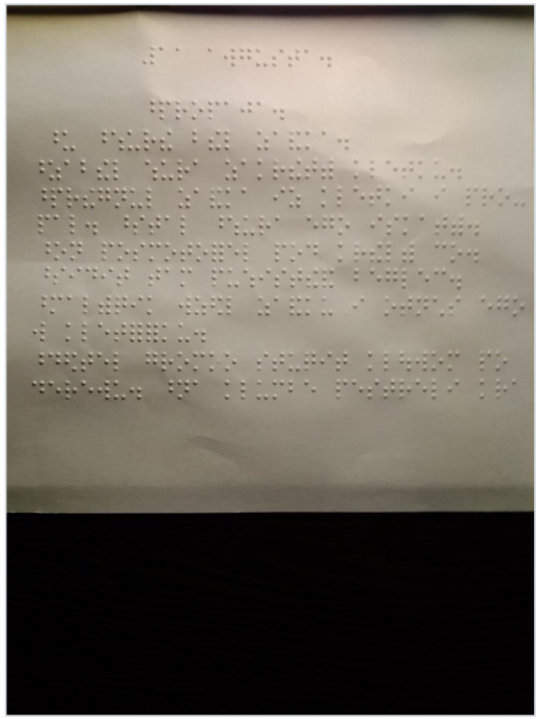
Book page

Angelina · 212



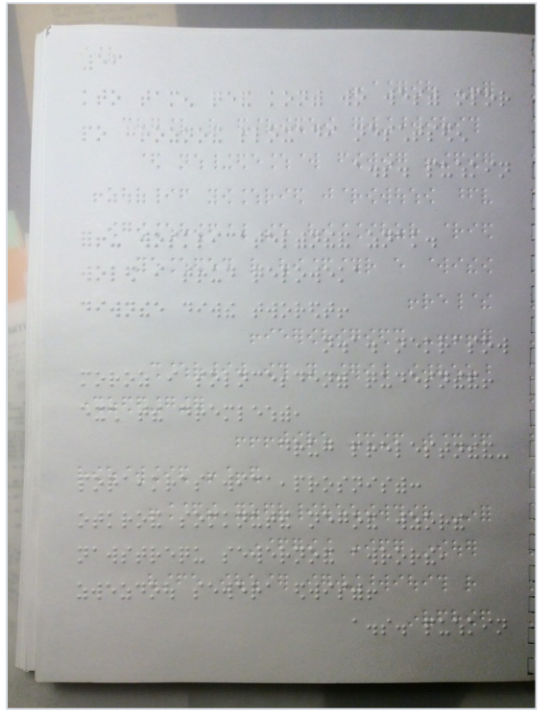
Handwritten

Angelina · 28



User upload

Angelina test · 50

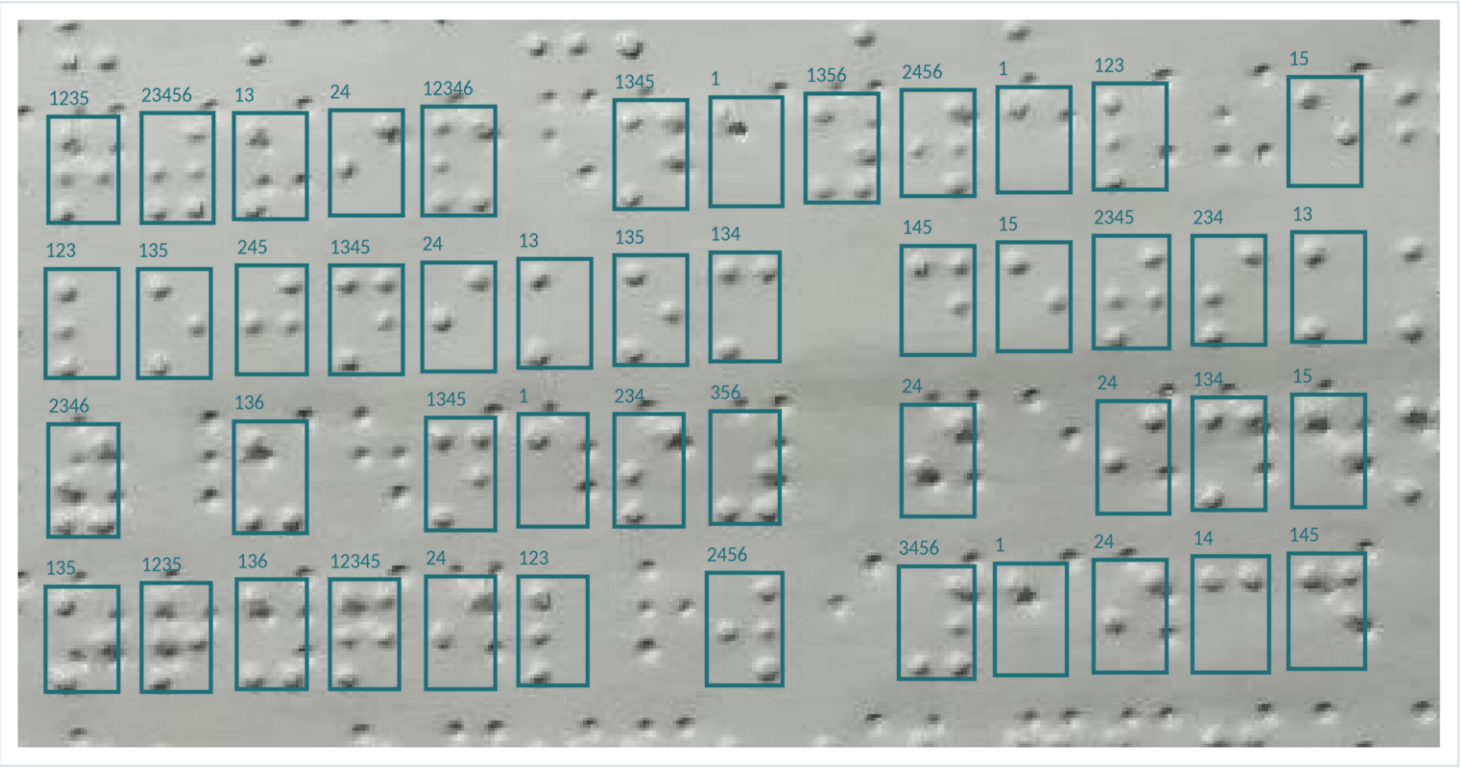


English book

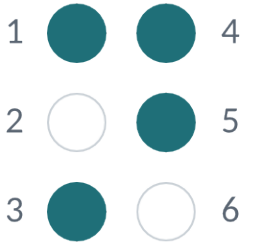
Book photos · 1,731

Loaded and rendered by our own script (code/explore.py). Not shown: 44 Angelina negatives (ordinary printed pages).

Ground truth: one box and one dot code per cell



Real annotation, drawn by our code: each box carries its dot code



Dots 1, 3, 4, 5 = “n”

1 + 4 + 8 + 16 = 29

dot i present → add 2^(i−1)

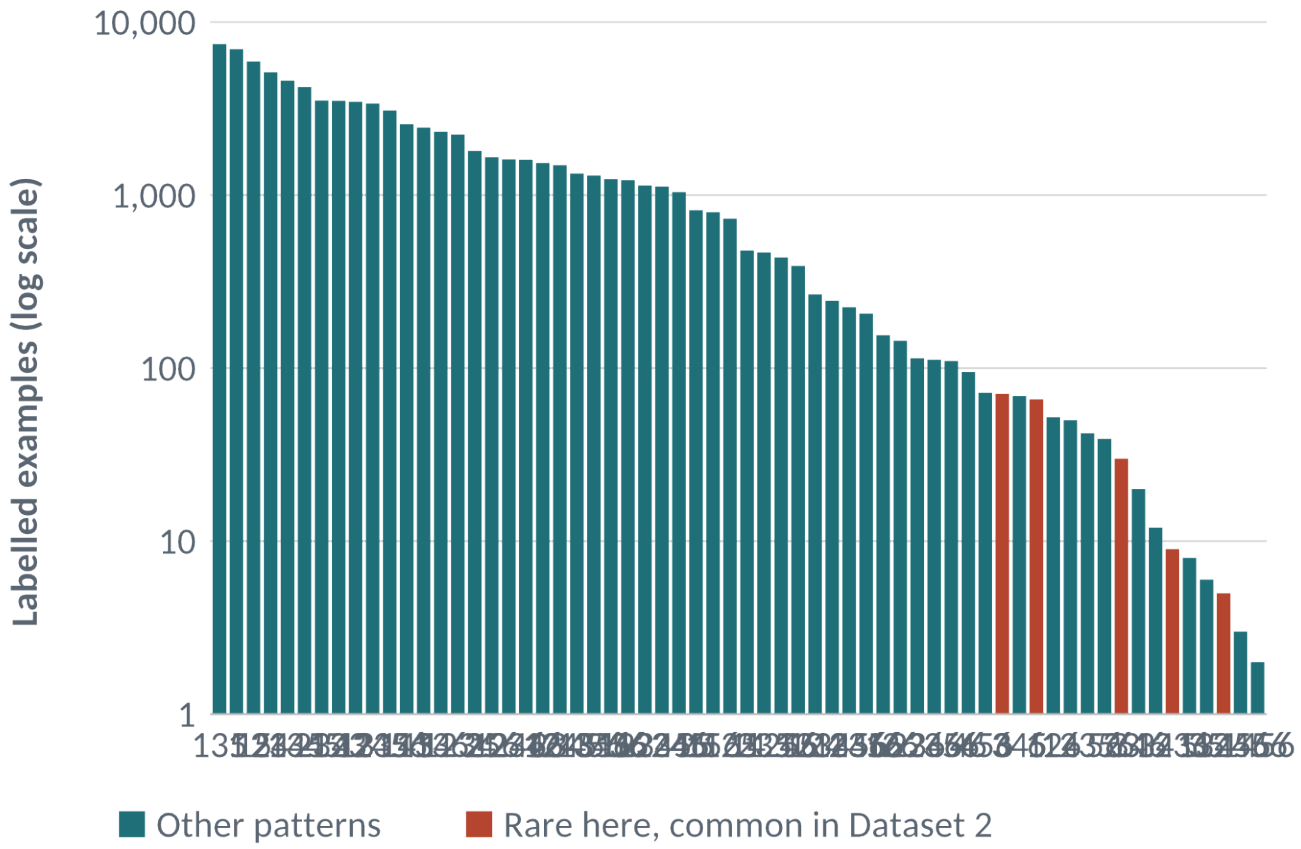
```
0.371;0.238;0.392;0.261;29
```

left; top; right; bottom (0–1); label

- One CSV per image
- Front side only is labelled

Class distribution: how often each dot pattern appears

All 85,263 labelled cells of Dataset 1, counted by our code



How to read it

- One bar = one dot pattern (a class)
- Height = number of examples
- Log scale: each grid line is 10×

Most common	Rarest
o (1-3-5): 7,472	th (1-4-5-6): 2
a (1): 6,973	er (1-2-4-5-6): 3
e (1-5): 5,918	y (1-3-4-5-6): 5

7,472 ÷ 2 = **3,736× spread**

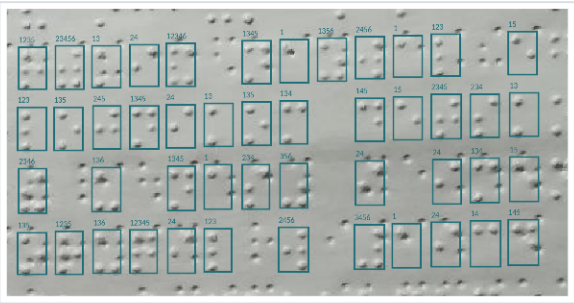
One real crop for every class



- Sorted from most to least frequent
- Each crop ≈ 22 × 33 px in the photo
- Rare classes are often faint or partly occluded
- Dots 1-4-6 never occur

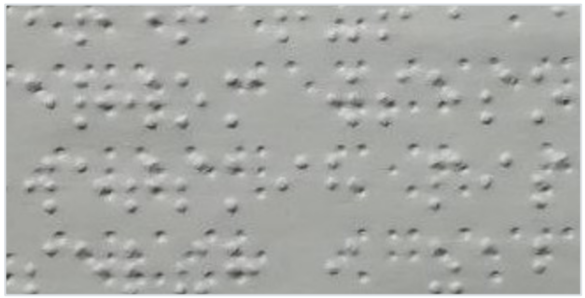
Crops cut from the dataset by code/explore.py; label = dot numbers (English letter when it has one), n = count

What makes the images hard, measured on the data



Tiny, dense objects

Median cell 22 × 33 px; 96 % are “small” by COCO rules



Unlabelled back side

212 book pages are double-sided; dents look like dots



train (book)



test (upload)

Blur and lighting

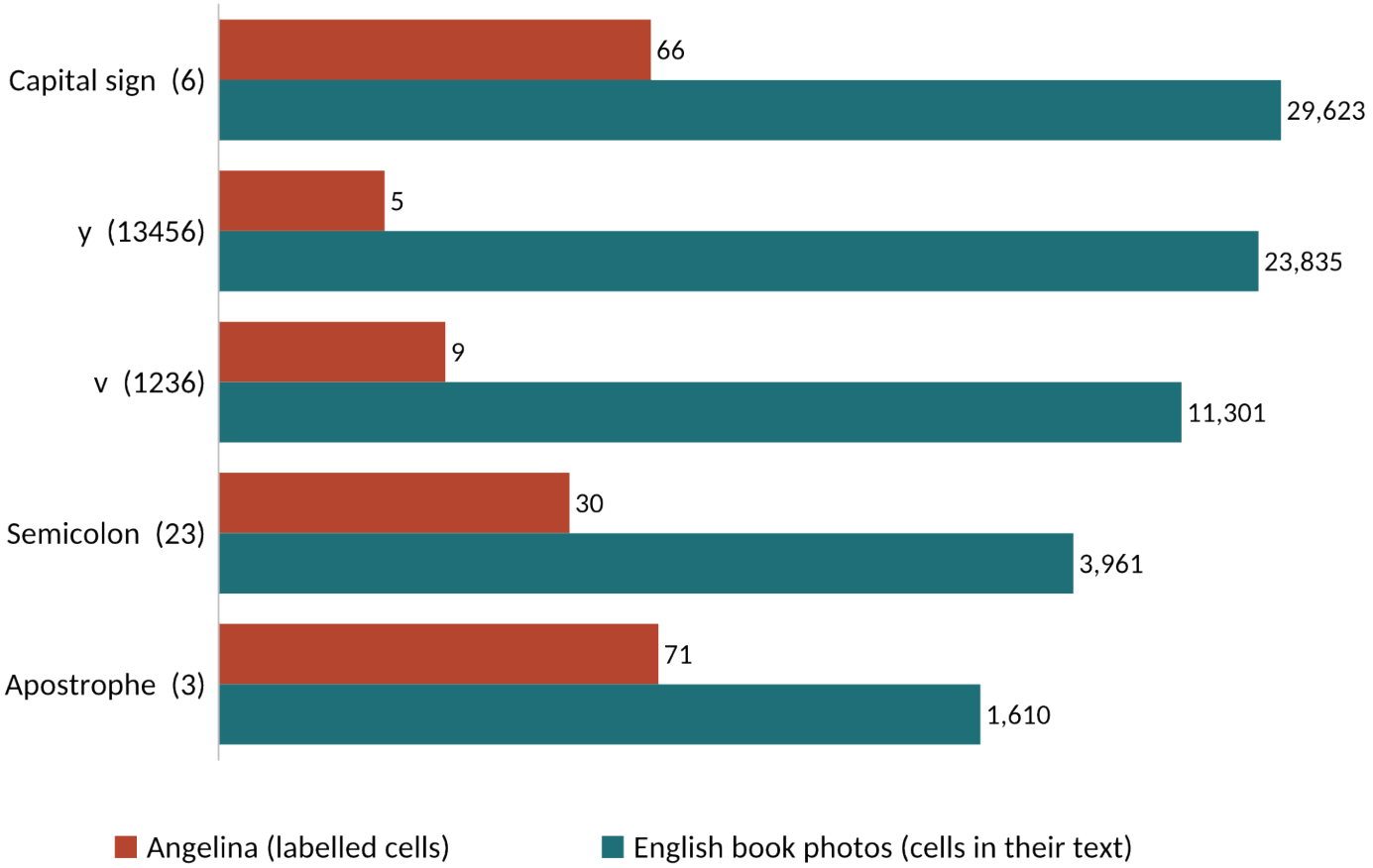
Test uploads are 2.7× blurrier than training books



Print inside Braille

Dual-format books: ink letters cross the Braille lines

The English book photos fill Angelina's rare classes



63 / 63

classes covered by the two sources together

0.84 → 0.996

word match after aligning labels to the book text (Zuev 2021)

Labels: a pretrained model reads each page, then the known text corrects it

English book counts: Gutenberg texts of both books translated to grade-1 English Braille with liblouis · Label correction: Zuev 2021

Why this data fits the task, and how we cover its limits

Why it fits

- Real phone photos: curved pages, tilt, uneven light
- Labels are dot patterns: any language via a table
- Per-cell boxes = exactly our output
- Published baseline to compare (char F1 0.998)

Limit → what we do

Few English cells in Angelina
→ add the English book photos

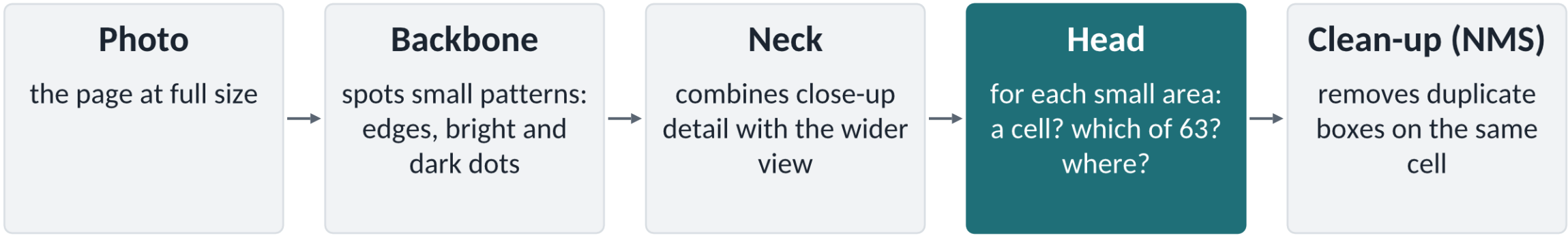
Book photos have text, not boxes
→ pseudo-label, then align to the text

Back side is unlabelled
→ learned as background

290 photos, 272 distinct pages
→ split by page, never by photo

English test: one book volume held out completely; its true text is known

Model: YOLO11-s finds every cell in one look



Why this model fits our data

- Finds hundreds of same-size cells in one look
- Cells never overlap: no separate “where to look first” step needed
- Small (9.4 M parameters): trains on one free GPU

What we compare it with

- RetinaNet: the published Braille model (Ovodov 2021)
- Same data, same scores → a fair comparison

YOLO11: Ultralytics 2024 (9.4 M parameters) · RetinaNet and focal loss: Lin et al., ICCV 2017 · Baseline: Ovodov 2021

Training: fine-tune, keep full resolution, augment safely

- **Fine-tune COCO weights, not from scratch**

Only 191 train pages, but 57,551 train cells

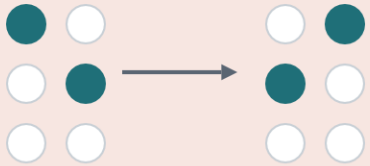
- **Keep native resolution (full page or 640 tiles)**

At a default 640 input a cell is ≈ 15 px, a dot ≈ 3.5 px

- **Train on Angelina + English book pages**

English labels: pseudo-labels corrected by the book text

Augmentations that corrupt labels



Horizontal flip turns “e” into “i”

180° rotation makes raised dots look like dents (front ↔ back)

Safe: small rotation, scale, blur, brightness

Tools and hardware

PyTorch | Ultralytics | OpenCV | Albumentations | liblouis | Colab / Kaggle T4 (16 GB)

Evaluation: detection metrics, then reading metrics

Metric	Why we report it
mAP@0.5	Main detection score: found and correctly classified
mAP@0.5:0.95	Box tightness — reported, but secondary for reading
Macro-F1 (63 classes)	Rare cells count as much as “o”
Character P / R / F1	Ovodov’s protocol (IoU ≥ 0.5, right class) → vs 0.998
Character error rate	English book pages: the true text is known

Test sets: Angelina user uploads (50 photos) · held-out English book volume

Not accuracy alone: predicting only the top 10 classes already covers 56 % of cells



Thank you — Questions?

- Task: detect and classify 63 dot patterns; English via the UEB table
- Data: 85,263 labelled cells + 1,731 English book photos; all 63 classes covered
- Next: fine-tune YOLO11-s at native resolution; mAP, macro-F1, character F1